

Customer Shopping Behavior Analysis Project

Project Overview

The Customer Shopping Behavior Analysis Project focuses on analyzing customer purchasing patterns, shopping trends, subscription behavior, and revenue insights using Python, SQL, and Power BI.

The project was developed as an end-to-end Data Analytics workflow that includes:

- Data loading and cleaning using Python
- Exploratory Data Analysis (EDA)
- SQL query analysis
- Interactive Power BI dashboard creation
- Reporting and visualization of customer behavior insights

This project demonstrates practical data analytics skills required for Data Analyst roles.

Dataset Information

The dataset contains customer shopping behavior data including:

- Customer demographics
- Product categories
- Purchase amounts
- Subscription status
- Review ratings
- Shipping preferences
- Age groups
- Shopping locations

The dataset was analyzed to identify customer purchasing trends and business insights.

Tools & Technologies Used

Programming & Analysis

- Python
- Pandas
- Jupyter Notebook

Database & Querying

- SQL
- PostgreSQL / MySQL
- pgAdmin

Data Visualization

- Power BI

Documentation & Presentation

- GitHub
 - PowerPoint / Gamma AI
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Project Workflow

1. Data Collection

Imported the customer shopping behavior dataset into Python using Pandas.

2. Data Cleaning

Performed:

- Missing value handling
- Data formatting
- Data type corrections
- Duplicate checking
- Column validation

3. Exploratory Data Analysis (EDA)

Analyzed:

- Customer purchase trends
- Product category performance
- Subscription behavior
- Shipping preferences
- Revenue distribution
- Age-group-based sales

4. SQL Analysis

Used SQL queries to:

- Filter customer records
- Aggregate sales data
- Analyze customer behavior
- Perform grouping and sorting
- Calculate revenue metrics
- Generate analytical insights

5. Dashboard Development

Built an interactive Power BI dashboard with:

- KPI cards
- Category analysis
- Revenue analysis
- Subscription analysis
- Sales by age group
- Revenue by age group
- Interactive slicers and filters

SQL Analysis

The SQL portion of the project includes:

- SELECT statements
- WHERE filtering
- GROUP BY & HAVING
- Aggregate functions
- ORDER BY
- CASE statements
- Data filtering
- Revenue calculations
- Customer segmentation

The SQL queries were used to extract meaningful customer insights and support dashboard reporting.

Power BI Dashboard Features

KPI Cards

- Total Customers
- Average Purchase Amount
- Average Review Rating

Interactive Filters

- Subscription Status
- Gender
- Category
- Shipping Type

Visualizations

- Revenue by Category
- Sales by Category
- Sales by Age Group
- Revenue by Age Group
- Subscription Distribution

The dashboard provides a clean and interactive business view of customer purchasing behavior.

Key Insights

- Clothing category generated the highest revenue and sales.
 - Most customers were non-subscribers.
 - Young adults contributed significantly to revenue generation.
 - Accessories and clothing were the most purchased product categories.
 - Standard and express shipping methods were commonly selected by customers.
 - Customer review ratings maintained a positive average score.
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Project Outcome

This project successfully demonstrates:

- End-to-end data analytics workflow
- SQL query writing skills
- Data cleaning and transformation
- Dashboard development skills

- Business insight generation
- Analytical thinking

The project strengthened practical knowledge in Python, SQL, and Power BI while building a professional analytics portfolio project.

Skills Demonstrated

- Data Cleaning
 - Exploratory Data Analysis
 - SQL Querying
 - Dashboard Design
 - Data Visualization
 - Business Analysis
 - Analytical Thinking
 - Problem Solving
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How to Run the Project

Python Analysis

1. Open Jupyter Notebook
2. Load the dataset
3. Run the Python notebook cells

SQL Queries

1. Open pgAdmin or MySQL
2. Import the dataset
3. Execute the SQL query file

Power BI Dashboard

1. Open the `.pbix` file
 2. Refresh dataset connections if required
 3. Explore dashboard visuals and filters
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Conclusion

The Customer Shopping Behavior Analysis Project provides a complete real-world data analytics solution using Python, SQL, and Power BI. The project demonstrates strong foundational and analytical skills suitable for Data Analyst and Business Intelligence roles.